CodingChallenge4_Markdown

Hikmat Paudel

2026-02-27

Contents

Noel et al. 2022, Plant Disease Paper

my Github repository: https://github.com/Hikmat-au/Assignments-repository

1 Question 1

YAML The YAML header is the section at the top of the R Markdown file enclosed by —. It contains data about title, author name, date of markdown creation, output format and table of contents.

Literate Programming Literate programming is a style where you write code and explanations together in the same document.
This helps readers understand why we wrote the code, not just what the code does.

2 Question2a

paperlink: (https://doi.org/10.1094/PDIS-06-21-1253-RE)

3 Question2b

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2

## Warning: package 'ggplot2' was built under R version 4.5.2

## Warning: package 'tibble' was built under R version 4.5.2

## Warning: package 'tidyr' was built under R version 4.5.2

## Warning: package 'readr' was built under R version 4.5.2

## Warning: package 'purrr' was built under R version 4.5.2

## Warning: package 'dplyr' was built under R version 4.5.2

## Warning: package 'stringr' was built under R version 4.5.2

## Warning: package 'forcats' was built under R version 4.5.2

## Warning: package 'lubridate' was built under R version 4.5.2

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(ggpubr)
```

```
## Warning: package 'ggpubr' was built under R version 4.5.2
```

```r
mycotoxin<- read.csv("MycotoxinData.csv", na.strings="na")
mycotoxin$Treatment <- as.factor(mycotoxin$Treatment)
mycotoxin$Cultivar <- as.factor(mycotoxin$Cultivar)
mycotoxin$DON <- as.numeric(mycotoxin$DON)
mycotoxin$X15ADON <- as.numeric(mycotoxin$X15ADON)
mycotoxin$MassperSeed_mg <- as.numeric(mycotoxin$MassperSeed_mg)
str(mycotoxin)
```

```
## 'data.frame':	375 obs. of  6 variables:
##  $ Treatment     : Factor w/ 5 levels "Fg","Fg + 37",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ Cultivar      : Factor w/ 2 levels "Ambassador","Wheaton": 2 2 2 2 2 2 2 2 2 2 ...
##  $ BioRep        : int  2 2 2 2 2 2 2 2 2 3 ...
##  $ MassperSeed_mg: num  10.29 12.8 2.85 6.5 10.18 ...
##  $ DON           : num  107.3 32.6 416 211.9 124 ...
##  $ X15ADON       : num  3 0.85 3.5 3.1 4.8 3.3 6.9 2.9 2.1 0.71 ...
```

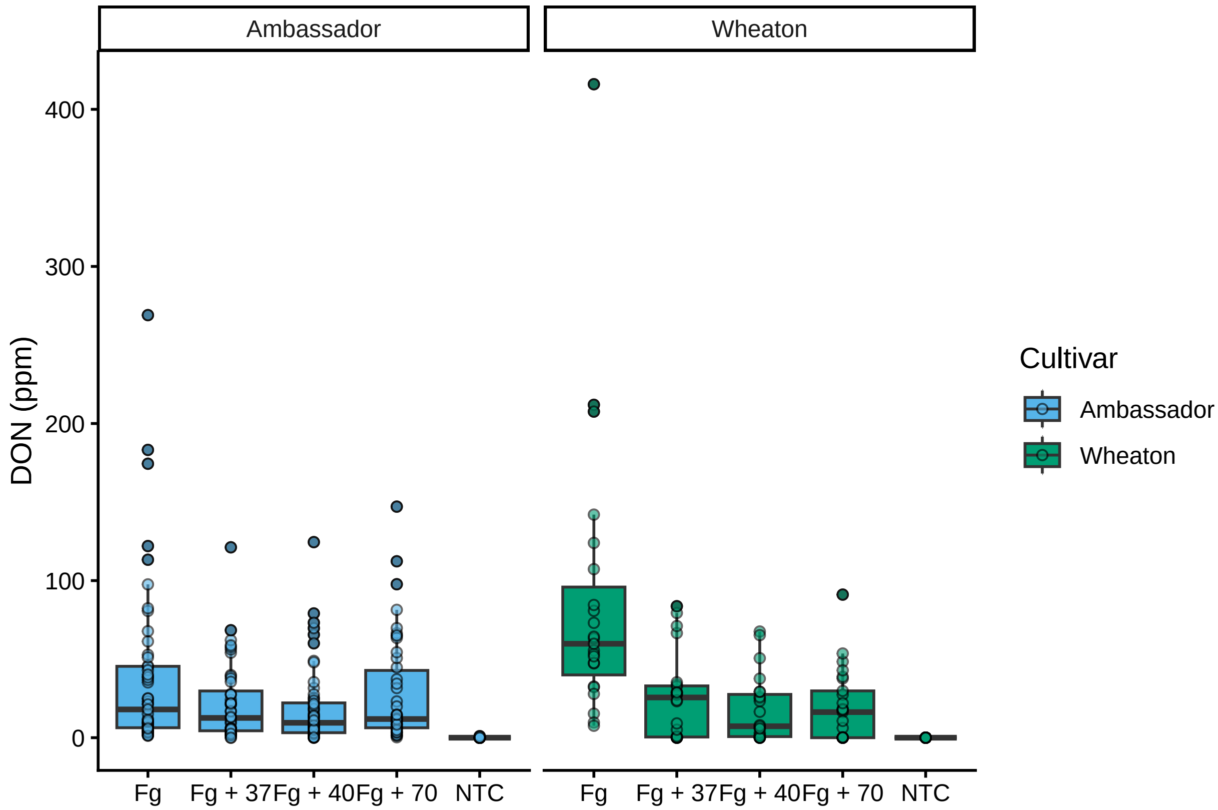
4 Question2c

```r
cbbPalette <- c("#56B4E9", "#009E73")
```

5 DON_PLOT1

```r
Plot1 <- ggplot(mycotoxin, aes(x = Treatment, y = DON, fill = Cultivar)) +
  geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("DON (ppm)") +
  geom_point(alpha = 0.6,pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette)+
  facet_wrap(~Cultivar)+
  theme_classic()
Plot1
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

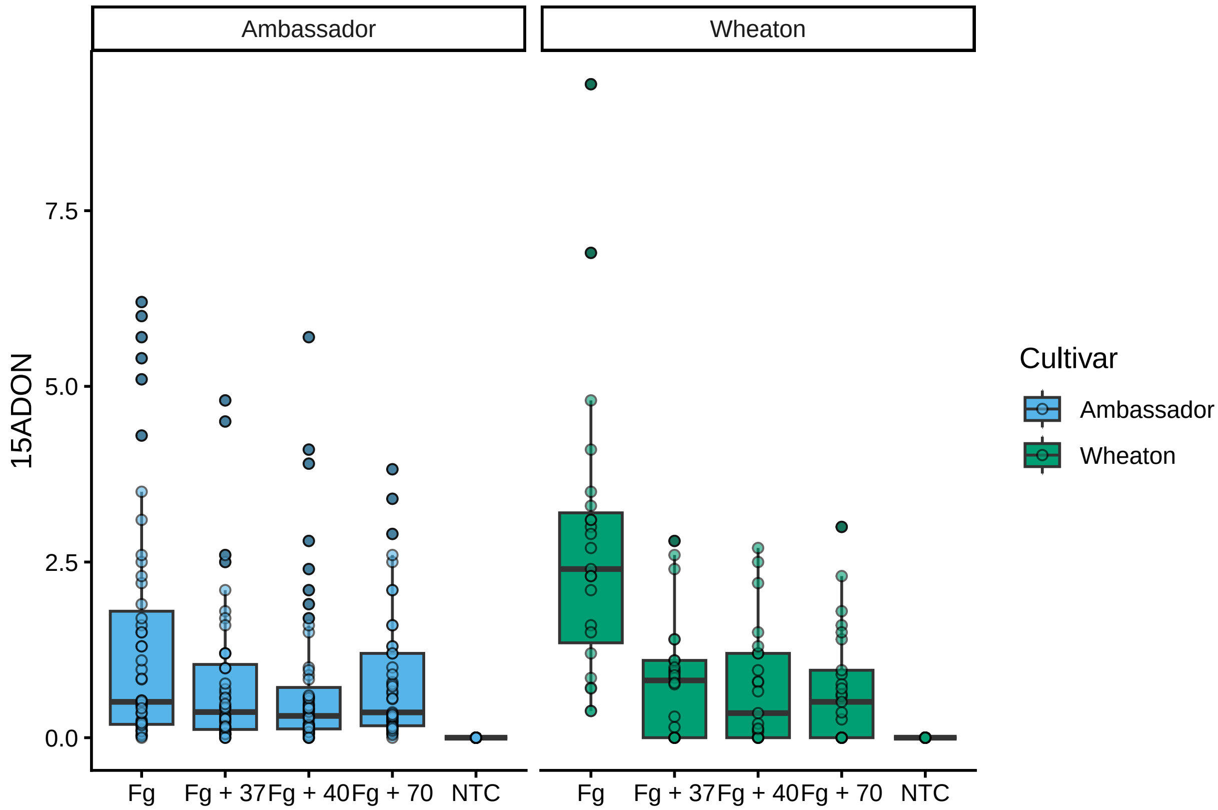


6 Plot X15ADON

```
Plot2 <- ggplot(mycotoxin, aes(x = Treatment, y = X15ADON, fill = Cultivar)) +
  geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("15ADON") +
  geom_point(alpha = 0.6,pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette)+
  facet_wrap(~Cultivar)+
  theme_classic()
Plot2
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

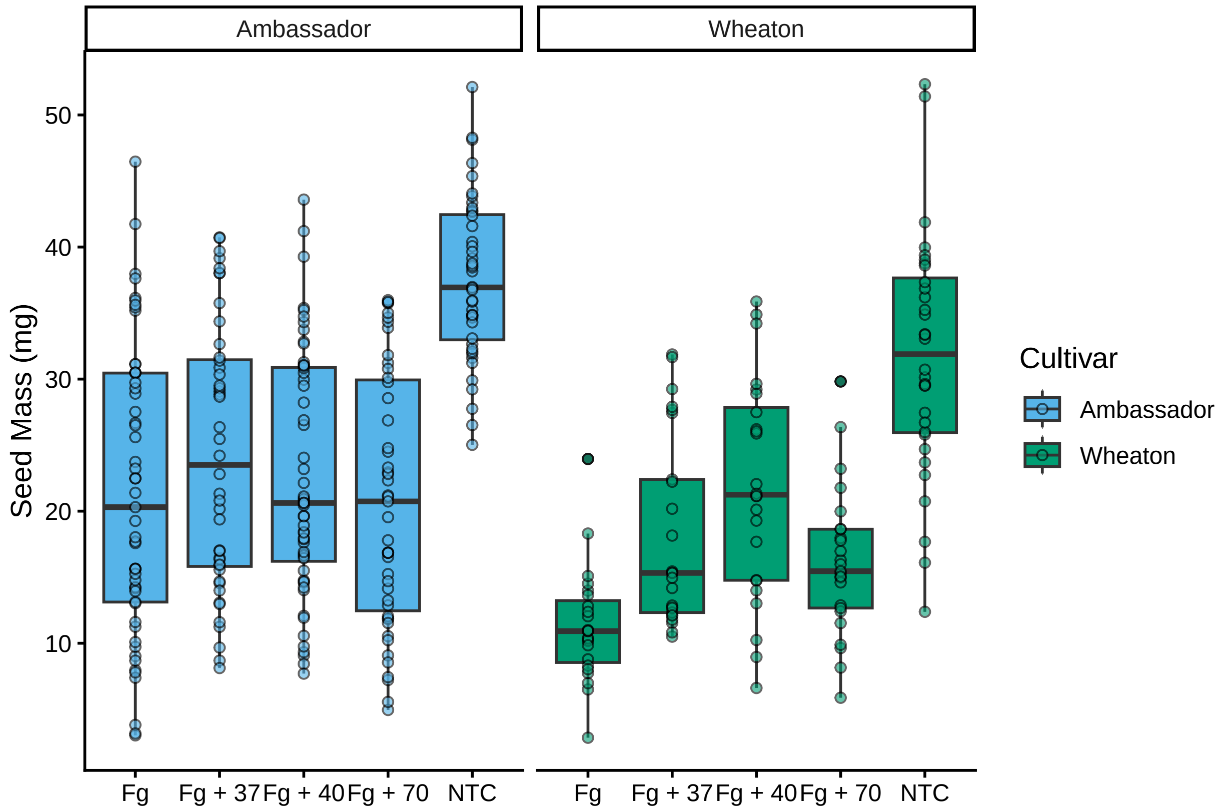


7 SeedMass

```
#Plot 3

Plot3<- ggplot(mycotoxin, aes(x = Treatment, y = MassperSeed_mg, fill = Cultivar)) +
geom_boxplot(position = position_dodge(0.85)) +
  xlab("") +
  ylab("Seed Mass (mg)") +
  geom_point(alpha = 0.6,pch = 21, color = "black") +
  scale_fill_manual(values = cbbPalette)+
  facet_wrap(~Cultivar)+
  theme_classic()
Plot3
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



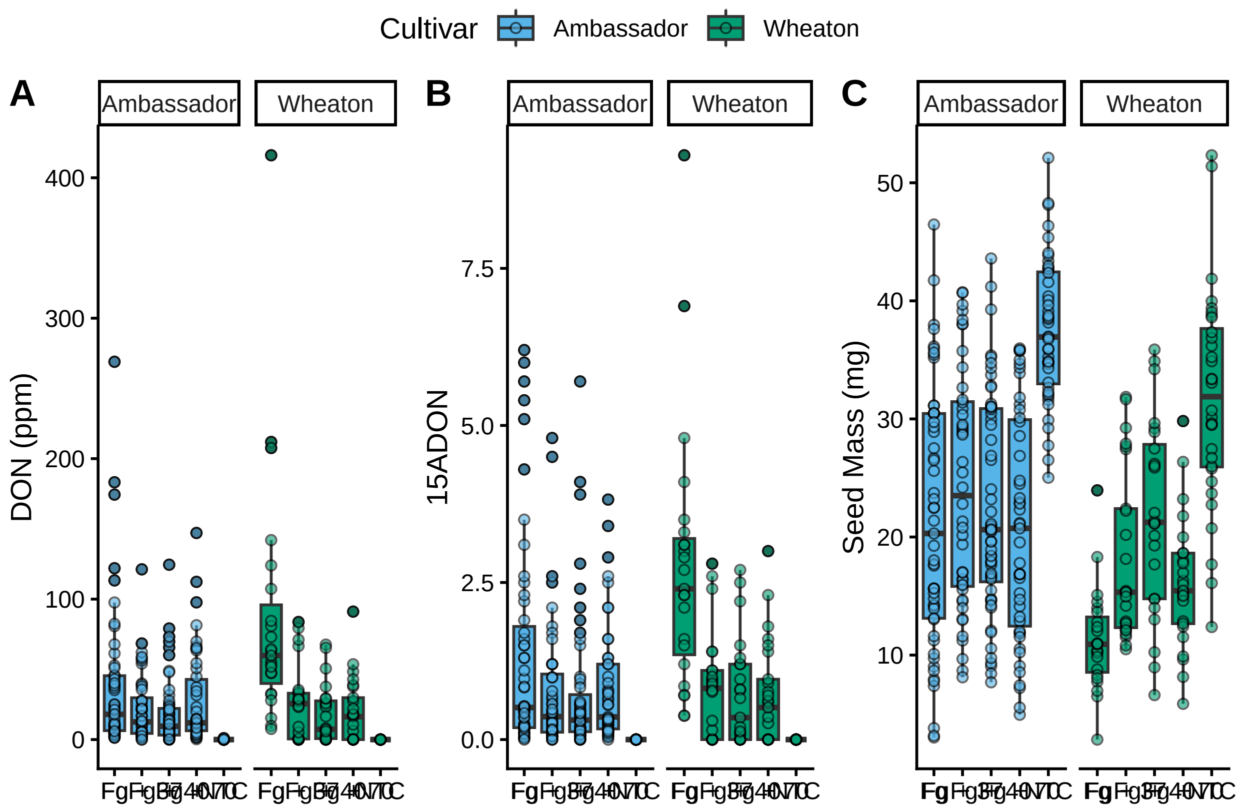
#3Combined_PLOT

```r
library(ggpubr)


figure=ggarrange(Plot1,Plot2,Plot3,labels = c("A", "B", "C"),nrow = 1,ncol = 3,common.legend=TRUE)
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```

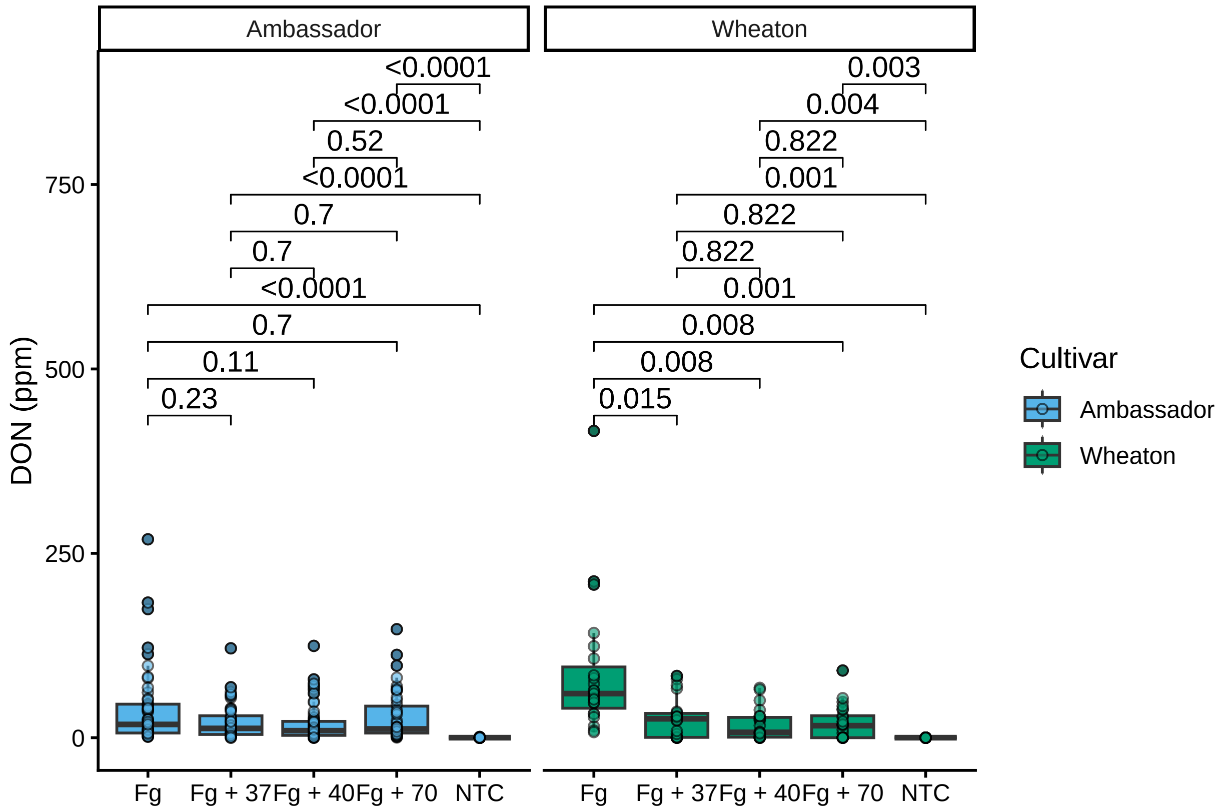
```
figure
```



#Combined_with_pairwisecomparison

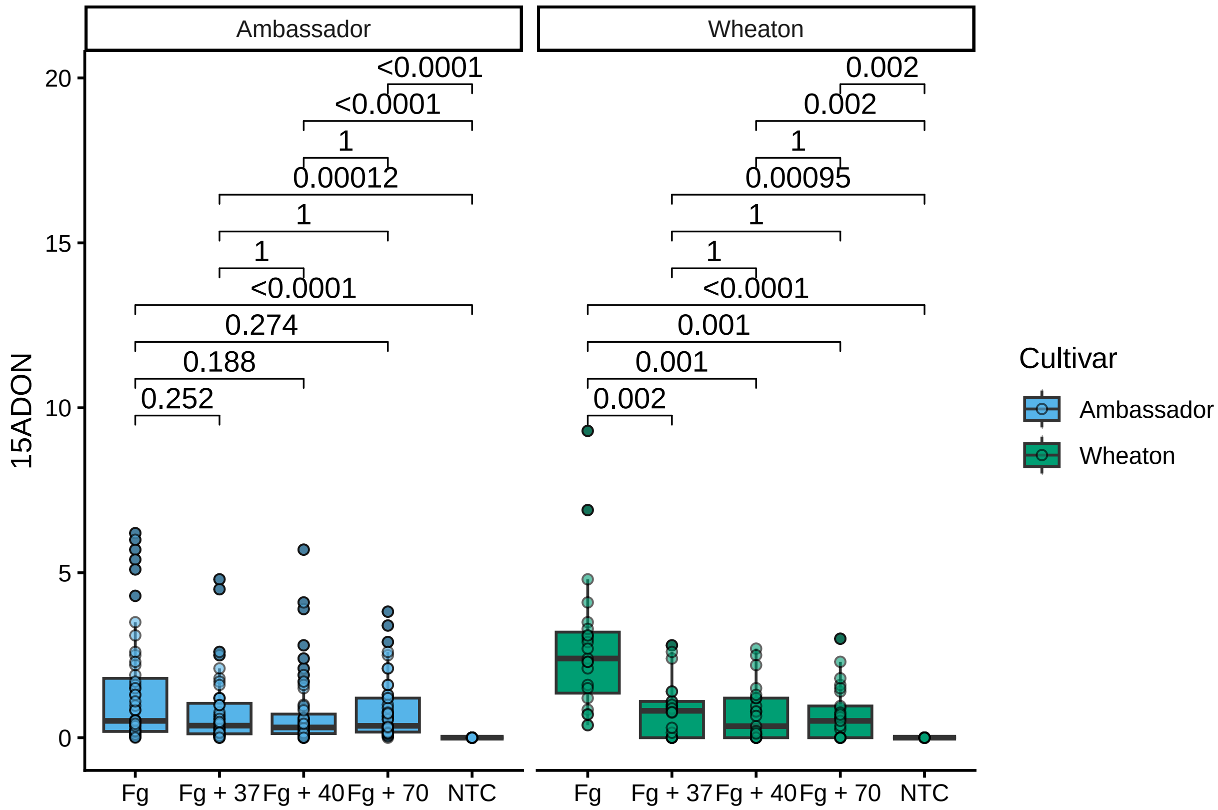
```
## Plot 1: plot1 with pairwise comparisons
Plot1_pwc <- Plot1 +
  geom_pwc(aes(group = Treatment), method = "t.test", label = "p.adj.format") +  # Add pairwise t-tests
  theme_classic()

Plot1_pwc
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).

## Warning: Removed 8 rows containing non-finite outside the scale range
## (‘stat_pwc()‘).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```




```r
##Plot 2: plot2 with pairwise comparisons
Plot2_pwc <- Plot2 +
  geom_pwc(aes(group=Treatment), method= "t.test", label= "p.adj.format")+
  theme_classic()

Plot2_pwc
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_pwc()`).

## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

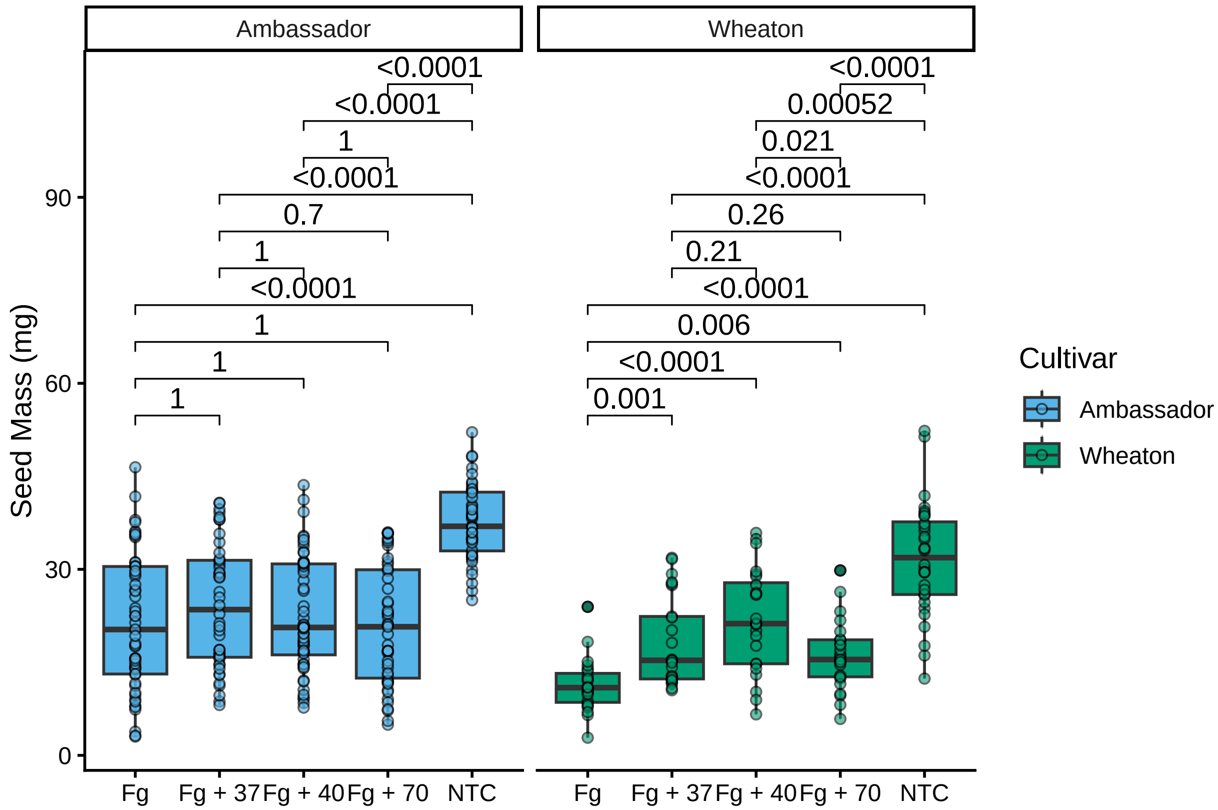



```r
##Plot 3: plot3 with pairwise comparisons
Plot3_pwc = Plot3 + geom_pwc(aes(group=Treatment), method="t.test", label="p.adj.format")+
  theme_classic()

Plot3_pwc
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_pwc()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

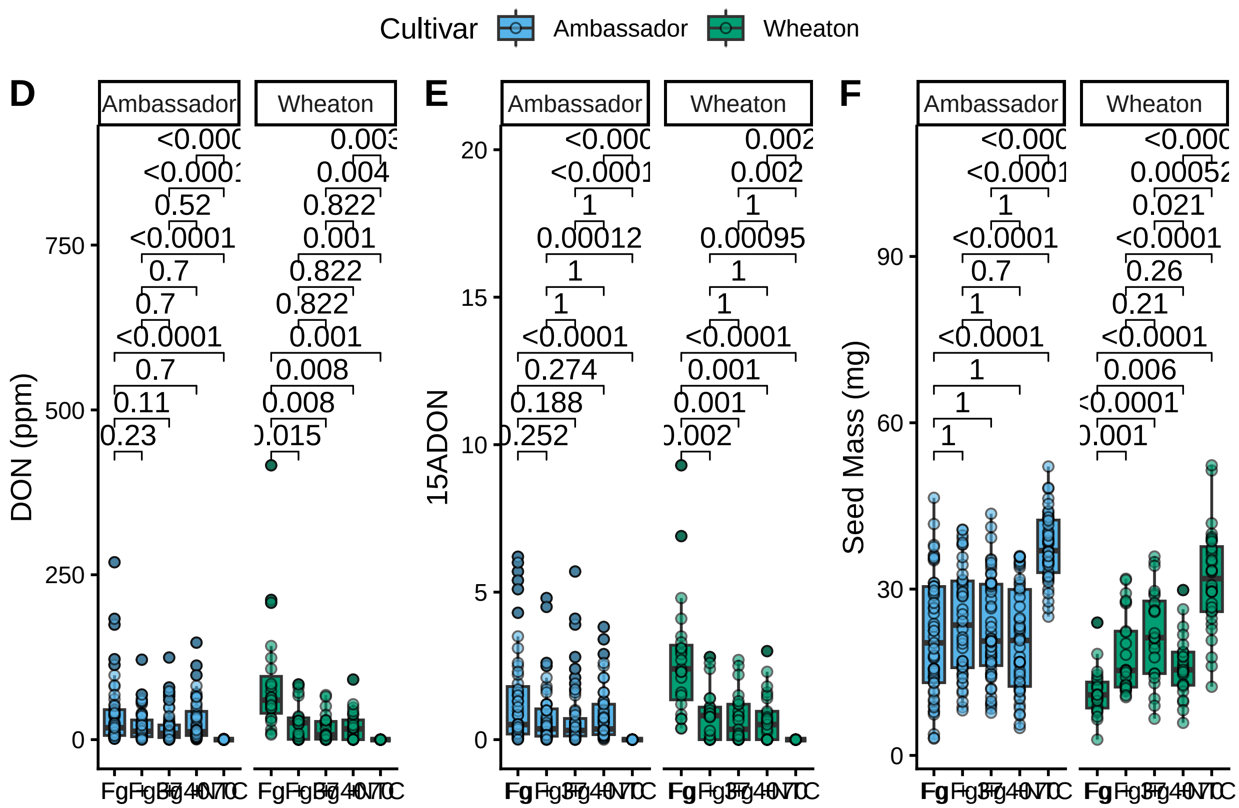



```
#Combine all three plots with a common legend ----
Combined_pwc =ggarrange(Plot1_pwc,Plot2_pwc,Plot3_pwc,labels = c("D", "E", "F"),nrow = 1,ncol = 3,commo
```

```
## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_pwc()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 8 rows containing non-finite outside the scale range
## (`stat_pwc()`).

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_pwc()`).

## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_pwc()`).

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

```
Combined_pwc
```



8 Question5

README

9 Question 6

Github my Github repository: https://github.com/Hikmat-au/Assignments-repository